

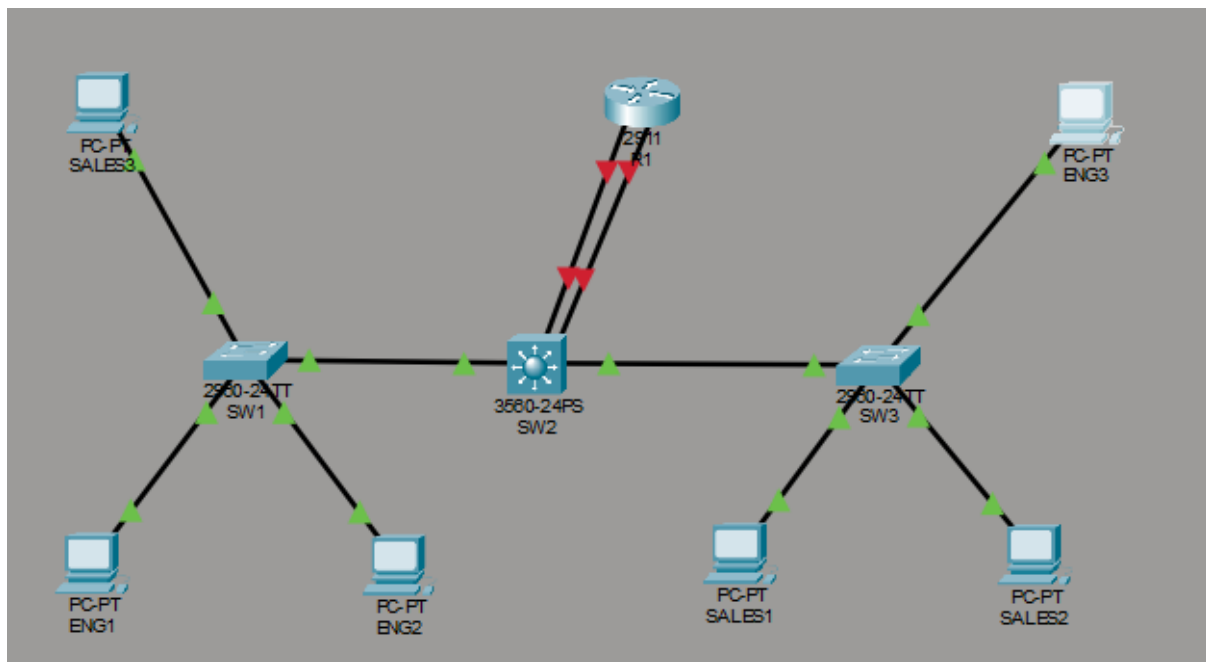


NET512 WEEK 9

Lerato Modise



SCREENSHOT OF A TOPOLOGY



Configure VTP, VLANs and switch ports

SW1

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vtp domain YOURDOMAIN
Changing VTP domain name from NULL to YOURDOMAIN
Switch(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to YOURDOMAIN.
vtp mode server
Device mode already VTP SERVER.
Switch(config)#vtp password YOURPASSWORD
Setting device VLAN database password to YOURPASSWORD
Switch(config)#
```

SW2

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vtp domain YOURDOMAIN
Changing VTP domain name from NULL to YOURDOMAIN
Switch(config)#vtp mode client
Setting device to VTP CLIENT mode.
Switch(config)#vtp password YOURPASSWORD%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain
name changed to YOURDOMAIN.
```

SW1

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)#name MGMT
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 20
Switch(config-vlan)#name STAFF
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 30
Switch(config-vlan)#name STUDENTS
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 40
Switch(config-vlan)#name GUEST
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 99
Switch(config-vlan)#name NATIVE
Switch(config-vlan)#exit
Switch(config)#
```

Show vlan brief

```
Switch>show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch>
```

Assign access ports to VLANs

SW1

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fa0/1-10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface range fa0/11-20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#
```

SW2

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fa0/1-10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#interface range fa0/11-20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#exit
Switch(config)#
```

Configure the trunk between SW1 and SW2

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,99
Switch(config-if)#no shutdown
Switch(config-if)#exit
```

```
Switch(config)#interface fa0/24
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,99
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	MGMT	active	
20	STAFF	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
30	STUDENTS	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20
40	GUEST	active	
99	NATIVE	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

```
C:\>ping 10.10.30.10
```

```
Pinging 10.10.30.10 with 32 bytes of data:
```

```
Reply from 10.10.30.10: bytes=32 time=1ms TTL=128
Reply from 10.10.30.10: bytes=32 time=3ms TTL=128
Reply from 10.10.30.10: bytes=32 time=2ms TTL=128
Reply from 10.10.30.10: bytes=32 time=2ms TTL=128
```

```
Ping statistics for 10.10.30.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms
```

```
Router(config)#interface g0/0.40
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 10.10.40.1 255.255.255.0
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state
to up

Router(config)#interface g0/0.99
Router(config-subif)#encapsulation dot1Q 99 native
Router(config-subif)#ip address 10.10.99.1 255.255.255.0
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.99, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state
to up

Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       10.10.10.1      YES manual up          up
GigabitEthernet0/0.10    unassigned      YES unset  up          up
GigabitEthernet0/0.20    unassigned      YES unset  up          up
GigabitEthernet0/0.30    unassigned      YES unset  up          up
GigabitEthernet0/0.40    10.10.40.1      YES manual up          up
GigabitEthernet0/0.99    10.10.99.1      YES manual up          up
GigabitEthernet0/1       10.10.20.1      YES manual up          up
GigabitEthernet0/2       10.10.30.1      YES manual down        down
Vlan1                    unassigned      YES unset  administratively down down
```

SW3

Physical Config CLI Attributes

```
Switch(config)#interface vlan 30
Switch(config-if)#ip address 10.10.30.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
%LINK-3-CHANGED: Interface Vlan30, changed state to up

Switch(config)#interface vlan 40
Switch(config-if)#ip address 10.10.40.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
%LINK-3-CHANGED: Interface Vlan40, changed state to up

Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/1          unassigned      YES manual up          up
FastEthernet0/2          unassigned      YES manual up          up
FastEthernet0/3          unassigned      YES manual up          up
FastEthernet0/4          unassigned      YES manual down        down
FastEthernet0/5          unassigned      YES manual down        down
FastEthernet0/6          unassigned      YES manual down        down
FastEthernet0/7          unassigned      YES manual down        down
FastEthernet0/8          unassigned      YES manual down        down
FastEthernet0/9          unassigned      YES manual down        down
FastEthernet0/10         unassigned      YES manual down        down
FastEthernet0/11         unassigned      YES manual down        down
FastEthernet0/12         unassigned      YES manual down        down
FastEthernet0/13         unassigned      YES manual down        down
FastEthernet0/14         unassigned      YES manual down        down
FastEthernet0/15         unassigned      YES manual down        down
FastEthernet0/16         unassigned      YES manual down        down
FastEthernet0/17         unassigned      YES manual down        down
FastEthernet0/18         unassigned      YES manual down        down
FastEthernet0/19         unassigned      YES manual down        down
FastEthernet0/20         unassigned      YES manual down        down
FastEthernet0/21         unassigned      YES manual down        down
```

```
C:\>ping 10.10.30.10

Pinging 10.10.30.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.
```

Reflection question

A Layer 3 switch is suitable for a large campus network because it can route traffic between many VLANs directly.